

PUNJABI UNIVERSITY PATIALA

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SUBJECT: DATA STRUCTURES

MST-2

MM:15

DATE: 14-1-2021

EXAMINERS: DR. DHAVLEESH RATTAN & DR.HARPREET KAUR

SECTION A IS COMPULSORY.

DO ANY 2 QUESTIONS FROM SECTION B

SECTION A

Q1:

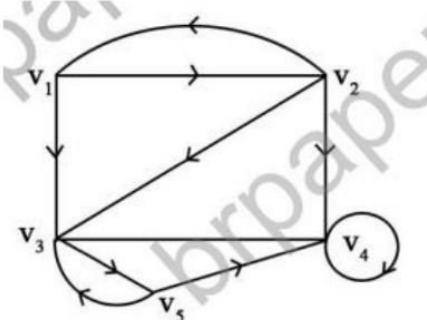
- How Trees are represented in memory?
- What do you mean by depth of tree?
- Write any two applications of graph.
- What is meant by strongly connected graph?
- What is breadth-first-search?

SECTION B

Q2: Write an algorithm for post-order tree traversal. Also show steps of algorithm on a set of numbers to show an example.

Q3: Build a heap H from the following list of numbers: 40, 65, 15, 48, 14, 50, 17, 22.

Q4: Consider the directed Graph G:



- Find indegree and outdegree of each node.
- Find number of simple paths.
- Is there any source or sink?

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Data Structures(2nd year)

Section –A(do all)

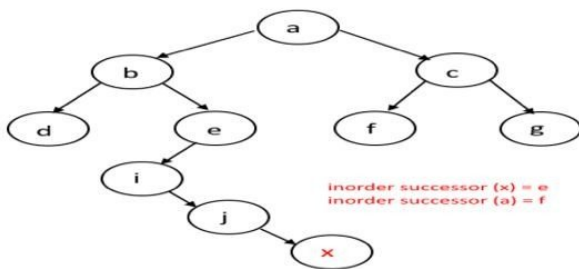
MST 2(All groups)

Q1) Can a Complete Binary tree and Full Binary tree be Perfect binary trees or vice versa justify your answer.....2)

Q2) What are the Maximum no of nodes in a Binary tree of height k?....1)

Q3) Consider a Graph G with 4 Vertices the Graph G will have How many Spanning trees are possible...1)

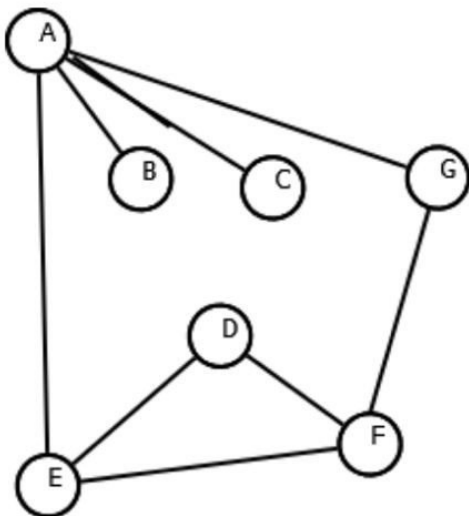
Q4) What is Inorder Successor of x in given BST.....1)



Section B(do any 2)

Q5) What is Heapify?How heap sort works? Explain with the help of example and algorithm....5)

Q6) Traverse a given Graph using BFS and use A as source node 5)



Q7) What is Complexity of deletion in Binary search Tree?Write an algorithm for deletion in BST.....5

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MST-2

B.Tech-3rd Sem (CSE)

MM: 15

Arnab Gang

1180

Data Structure (CSEB2302T)

Time: 1 hour

Note: Q1 is Compulsory. Attempt any two questions from Q2, Q3 and Q4. Each question carries five marks.

Q1. Explain the purpose of following (1x5=5 Marks)

i) Directed Graph ii) Complete Binary Tree iii) File Organization iv) Rotations in AVL Tree v) Adjacency Matrix

Q2. How Insertion, deletion and searching takes place in Binary search tree? Explain with suitable example.

Q3. How Heap sorting works? Explain with suitable example.

Q4. How Breadth First search and Depth first search works in a graph? Explain with suitable example.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PUNJABI UNIVERSITY, PATIALA

MST – 2

DATA STRUCTURES (CSEB2302T)

B.Tech. – 2nd Year (3rd Sem)

Time: 1 Hour

Max Marks : 15

4th Nov, 2025 (Tuesday)

Section – A (Attempt ALL questions)

1. How to insert an element in a binary search tree? (2)
2. Working of breadth first search technique in graphs. (2)
3. Importance of adjacency matrix. (1)

Section – B (Attempt any TWO questions)

4. Discuss the Dijkstra's algorithm in detail. (5)
5. Explain heap sort algorithm with suitable example. (5)
6. Differentiate between sequential, relative and indexed sequential file organization techniques. (5)

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DATA STRUCTURES (MST - II)

Time Duration: 1 Hr.

Max Marks: 15

Note: All the Questions are compulsory in Section A. Attempt any two questions from Section B

Section-A

Q1. Define properties of Binary Tree.

(1)

Q2. How collision is resolved in hashing?

(2)

Q3. Construct a binary search tree from given elements:

7, 4, 12, 3, 6, 8, 1, 5, 10

(2)

Section-B

(2x5=10)

Q4. Discuss the sequential and linked representation of graph-adjacency matrix.

Q5. Differentiate between depth first search and breadth first search. Give suitable example.

Q6. Explain the various file organization techniques in detail.

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B.Tech-II (CE) CPE-208 Data Structures
MST-2

MM: 15

Time: 1 hour

Section-A (1*5=5)

- 1. Differentiate between linear and non linear data structures.
- 2. Differentiate between binary tree and binary search tree.
- 3. What are minimal spanning trees?
- 4. Explain the difference between max heap and min heap.
- 5. What is the benefit of using adjacency list over adjacency matrix?

Section-B(2*5=10)

- 1. Explain Dijkstra's algorithm in detail with the help of an example.
- 2. Write an algorithm to insert a node in the middle of a single linked list with the help of an example.
- 3. Explain various traversals in a binary search tree.

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B.Tech-II (CE) CPE-208 Data Structures Second Mid Semester Test

Time: 1 hour

MM: 15

Q1 Define Following Terms (1*5)

- a) Spanning Tree b) Graph c) PST d) Sequential files e) Relative files

Attempt any 2 questions (each question carries 5 marks)

1. Explain Heap And Heap Sort algorithm with help of example.
2. Explain various methods to represent Graphs.
3. Write algorithm to insert a node in Binary Search tree

Department of Computer Science & Engineering, Punjabi University, Patiala

MST-2
B.Tech-3rd Sem (CSE)

12101078 Prem Kumar

MM: 15
Time: 1 hour

Data Structure

Note: Q1 is Compulsory, each part carrying 1 mark. Attempt any two questions from Q2, Q3 and Q4, each question carrying 5 marks.

Q1. Define a) Preorder of $(a+b)-(c-d)$. b) Threaded tree c) Depth First Search d) Sequential file organization.
e) Rotation in an AVL Tree

Q2. Write down and explain Dijkstra Shortest Path Algorithm with the help of an example.

Q3. Explain in detail the insertion and deletion of a node in a Binary search tree with the help of an example

Q4. Explain in detail Representation of Binary trees in memory.